

Nova Electrical and Thermal Verification Report

Report scope

Requirements evaluated: 12

Domains: Electrical safety, Thermal control

Methods: bench test, environmental exposure, inspection, log review

Result vocabulary: PASS, FAIL, NOT TESTED, INCONCLUSIVE

1. Executed results

Each disposition applies only to the criterion stated in the same record.

REQ-NVA-ELE-001 Operating voltage window	PASS - Executed for REQ-NVA-ELE-001; the recorded result confirms: Functional coverage includes 350 V DC. PASS - Executed for REQ-NVA-ELE-001; the recorded result confirms: Functional coverage includes 820 V DC.	Bench / inspection record Reviewed by DV team
REQ-NVA-ELE-002 Sleep current	PASS - Executed for REQ-NVA-ELE-002; the recorded result confirms: Wake line remains low for at least 120 seconds. IN PROGRESS - Partial execution exists for REQ-NVA-ELE-002, but verification is not complete: Controller supply current does not exceed 5 mA.	Bench / inspection record Reviewed by DV team
REQ-NVA-ELE-003 Post-impact isolation	PASS - Executed for REQ-NVA-ELE-003; the recorded result confirms: Isolation measurement occurs 60 seconds after impact. FAIL - Executed for REQ-NVA-ELE-003; the observed result does not satisfy: Isolation resistance per bus voltage is at least 500 ohm/V.	Bench / inspection record Reviewed by DV team
REQ-NVA-ELE-004 Contactor opening	NOT TESTED - No test or inspection was performed for REQ-NVA-ELE-004: A validated crash input is received. NOT TESTED - No test or inspection was performed for REQ-NVA-ELE-004: Both positive and negative contactors reach open state. NOT TESTED - No test or inspection was performed for REQ-NVA-ELE-004: Opening completes within 20 ms.	Bench / inspection record Reviewed by DV team

2. Executed results

Each disposition applies only to the criterion stated in the same record.

REQ-NVA-ELE-005 Precharge behavior	INCONCLUSIVE - Work was attempted for REQ-NVA-ELE-005, but incomplete scope or calibration prevents deciding: DC link reaches at least 90 percent of pack voltage. INCONCLUSIVE - Work was attempted for REQ-NVA-ELE-005, but incomplete scope or calibration prevents deciding: Required level is reached within 2.0 seconds. INCONCLUSIVE - Work was attempted for REQ-NVA-ELE-005, but incomplete scope or calibration prevents deciding: DC-link overshoot above pack voltage does not exceed 10 V.	Bench / inspection record Reviewed by DV team
REQ-NVA-THM-001 Ambient operating range	PASS - Executed for REQ-NVA-THM-001; the recorded result confirms: Monitoring functions operate at -30 deg C. NOT TESTED - No test or inspection was performed for REQ-NVA-THM-001: Monitoring functions operate at +70 deg C.	Bench / inspection record Reviewed by DV team
REQ-NVA-THM-002 Shutdown band	PASS - Executed for REQ-NVA-THM-002; the recorded result confirms: Thermal shutdown is commanded at or above 90 deg C. FAIL - Executed for REQ-NVA-THM-002; the observed result does not satisfy: Thermal shutdown is not commanded below 85 deg C.	Bench / inspection record Reviewed by DV team
REQ-NVA-THM-003 Coolant delivery	NOT TESTED - No test or inspection was performed for REQ-NVA-THM-003: Battery heat rejection is 35 kW. NOT TESTED - No test or inspection was performed for REQ-NVA-THM-003: Pack coolant flow is at least 8.0 L/min.	Bench / inspection record Reviewed by DV team

3. Executed results

Each disposition applies only to the criterion stated in the same record.

REQ-NVA-THM-004 Controlled recovery	INCONCLUSIVE - Work was attempted for REQ-NVA-THM-004, but incomplete scope or calibration prevents deciding: Temperature is below 70 deg C. INCONCLUSIVE - Work was attempted for REQ-NVA-THM-004, but incomplete scope or calibration prevents deciding: Temperature remains below the limit for 60 seconds. INCONCLUSIVE - Work was attempted for REQ-NVA-THM-004, but incomplete scope or calibration prevents deciding: No latched temperature-sensor fault is present.	Bench / inspection record Reviewed by DV team
REQ-NVA-THM-005 Cell temperature uniformity	PASS - Executed for REQ-NVA-THM-005; the recorded result confirms: Discharge rate is 2C. PASS - Executed for REQ-NVA-THM-005; the recorded result confirms: Discharge duration is at least 30 minutes. PASS - Executed for REQ-NVA-THM-005; the recorded result confirms: Maximum cell-temperature difference is 5 deg C or less.	Bench / inspection record Reviewed by DV team

4. Image-based inspection result

Visual Inspection Record

REQ-NVA-ELE-006 | Emergency discharge

PASS

the recorded result confirms: Exposed DC bus falls below 60 V.

PASS

the recorded result confirms: Voltage limit is achieved within 5.0 seconds.

PASS

the recorded result confirms: Red high-voltage warning indicator illuminates.

REQ-NVA-THM-006 | Cooling flow direction

PASS

the recorded result confirms: Air moves from the blue inlet toward the red outlet.

UNTESTED

NOT TESTED - No test or inspection was performed for REQ-NVA-THM-006: Cooling airflow is at least 420 m3/h.

UNTESTED

NOT TESTED - No test or inspection was performed for REQ-NVA-THM-006: Service clearance is unobstructed.

Image record NOVA-DV-VIS | Reviewer: QA-17 | Controlled copy

Figure 4-1. Recorded inspections for REQ-NVA-ELE-006, REQ-NVA-THM-006; interpretation requires the raster content.